The empirical recurrence for OEIS A234892: recovering a conjecture that is not written down

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Abstract

OEIS A234892 records "Empirical recurrence of order 51", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 252$ vertices and satisfies some monic recurrence of order at most S; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 51, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 51 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A234892 is "Number of (n+1) X (2+1) 0..6 arrays with every 2 X 2 subblock having its diagonal sum differing from its antidiagonal sum by 1, with no adjacent elements equal (constant-stress tilted 1 X 1 tilings).". It has offset 1 and begins

$$760, 4024, 20404, 115924, 625480, 3640580, 19965444, 117487100, \ \ldots$$

The entry states:

> Empirical recurrence of order 51 (see link above).

As of the "Last modified" line on the live entry (Jun 20 2022 01:56:04, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 252$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S =$ 252.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 504$ exact terms of the model returns order 51 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{51} c_i \, a(n-i),$$

c_1	4	c_{14}	4498389253	c_{27}	−33116019498798	c_{40}	3987806160592
c_2	123	c_{15}	−18768892850	c_{28}	−1490305382094	c_{41}	6051932417256
c_3	−483	c_{16}	−26297853312	c_{29}	52810649104079	c_{42}	−1425570357572
c_4	−6531	c_{17}	116335244492	c_{30}	−2363703417987	c_{43}	−1588839760344
c_5	25275	c_{18}	117932927606	c_{31}	−67448269920756	c_{44}	347288324768
c_6	200925	c_{19}	−566371038157	c_{32}	7715763511555	c_{45}	288115053744
c_7	−772150	c_{20}	−405904133091	c_{33}	68635991346730	c_{46}	−53678726048
c_8	−4045683	c_{21}	2187138687608	c_{34}	−11455498172200	c_{47}	−33399316384
c_9	15587027	c_{22}	1061356877539	c_{35}	−55199111713794	c_{48}	4603347312
c_{10}	56995150	c_{23}	−6744275519606	c_{36}	11279581805386	c_{49}	2152678752
c_{11}	−222557768	c_{24}	−2044558418304	c_{37}	34670307025860	c_{50}	−159862848
c_{12}	−585318095	c_{25}	16674705188002	c_{38}	−7904850175210	c_{51}	−55125120
c_{13}	2345380349	c_{26}	2652227331231	c_{39}	−16724353625432		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 51, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $51+4$ terms removed returns order 51 as well, so $\deg q_{\min} = 51 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $51+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 51 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A234892.

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